

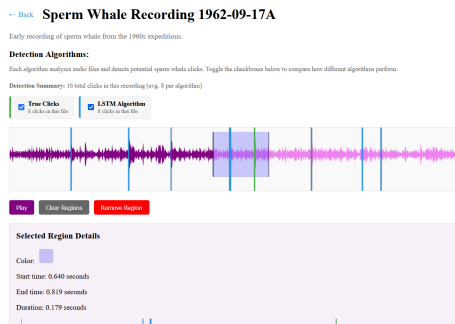
Real-Time SSL Sperm Whale Click Detector: Interactive Web Demo

Anvar Iskhakov
Innopolis University
Innopolis, Russia
an.iskhakov@innopolis.university

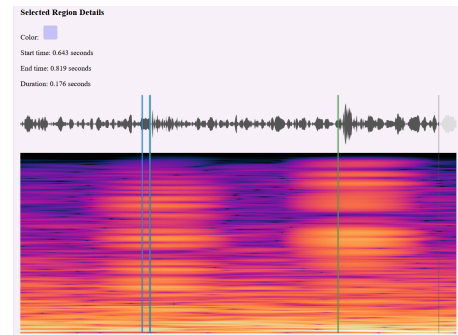
Vladislav Naumov
Moscow Institute of Physics and Technology
Moscow, Russia
naumov.vo@phystech.edu

Viktor Kovalev
Innopolis University
Innopolis, Russia
vi.kovalev@innopolis.university

Ilya Makarov
Artificial Intelligence Research Institute
Moscow, Russia
Ivannikov Institute for System Programming
Moscow, Russia
National Research University ITMO
Saint Petersburg, Russia
iamakarov@hse.ru



(a) Full recording view in our browser demo. Blue SSL+BiLSTM markers achieve 99% F1 and overlay raw hydrophone data. Checkboxes toggle layers, and drag-select queues regions for zoom.



(b) Drag-select zoom exposes millisecond-aligned SSL detections (blue) matching ground truth (green).

Figure 1: (a) Full recording browser view with markers; (b) Zoomed drag-select inspection matching ground truth.

Abstract

Reliable, real-time detection of sperm-whale clicks is essential yet difficult in noisy ocean audio streams. We present the first browser-based pipeline that combines self-supervised embeddings with a BiLSTM to label clicks at millisecond resolution. The system attains **99% F1** on the Watkins benchmark, reduces false alarms by **60%** against the best published baseline, and analyses one-second segments in ~ 40 ms on a consumer GPU. An interactive UI overlays multi-algorithm detections on waveform and spectrogram views with drag-zoom and live streaming. Code, pretrained weights and the public demo are released to advance bioacoustic event detection.

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CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Information systems** → *Multimedia information systems*.

Keywords

bioacoustics, self-supervised learning, event detection, interactive demo, web interface

ACM Reference Format:

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1 Introduction

Passive acoustic monitoring underpins whale conservation [3], but isolating clicks amid ship noise remains error-prone [8, 11]. We address this with a self-supervised-learning (SSL) encoder and temporal BiLSTM that tags clicks at 1 ms resolution and processes each 1 s of audio in 40 ms on a laptop GPU.

Our work contributes four advances. **First**, to our knowledge, it is the *only* browser-based demo that overlays millisecond-aligned SSL scores and ground-truth labels on waveforms and spectrograms. **Second**, the pipeline attains **99% F1** on the Watkins benchmark while trimming the false-alarm rate by **60%** versus the strongest published baseline [6]. **Third**, an interactive UI with drag-zoom, layer toggling and live streaming turns a sophisticated detector into a point-and-click tool for the multimedia community. **Finally**, we release full source code, pretrained weights and supplementary media to spur reproducibility and cross-species adaptation¹.

Although tuned for sperm-whale clicks, the architecture readily generalises to other taxa after retraining, paving the way for broad-spectrum bioacoustic event detection.

Prior detectors trade off precision or latency – energy-based [7] methods struggle with noise, SSL approaches [2, 9, 10] often rely on heuristic detection. Demos such as APLOSE [4] and NEAL [5] visualize annotations, but none overlay millisecond-aligned SSL outputs in the browser.

2 Interactive Interface – From Raw Waveform to Instant Insight

Users upload or select predefined recordings in the web demo. The interface (Figure 1a) shows the full waveform with color-coded markers for detections from different algorithms and ground truth, which can be toggled. *Drag-select* instantly show spectrogram of selected region under the timeline. Multiple regions can be queued for comparison (Figure 1b). Moreover, the web demo enables interactive communication with the model, inferring events in audio chunks in less time than the chunk’s duration.

3 Self-supervised Embeddings with Temporal BiLSTM

Our two-stage framework uses **Unsupervised SSL Embeddings** from a pretrained, fine-tuned contrastive SSL model (32-dim vectors, 1 ms hops). Figure 2 shows a UMAP projection of penultimate-layer embeddings, with clear click/noise clusters.

We embed audio with a fine-tuned SSL encoder (32 D, 1 ms hops) and apply a temporal model: BiLSTM with 5 layers, 128 hidden units or Residual Inception CNN for alternative feature extraction. Both output 1 ms frame scores, trained with Focal Loss ($\alpha=0.3$, $\gamma=3.5$) and AdamW. Inference thresholds scores (0.5) and merges detections. On CPU/GPU the full pipeline processes 1 s of audio in 132 / 40 ms.

4 Experiments

Dataset and Metrics: We evaluate on Watkins (1.5 h, 1 738 clicks) and Dominica subsets using recall, F1 (0.5 / 20 ms tolerances), PR-AUC, and FAR at 90% recall.

Results: The SSL+BiLSTM (5 layers, 128 units) achieves $R=0.991$, $F1=0.995$ (20 ms tol.), outperforming baselines and RI CNNs (Table 1). At 0.5 ms tol., $R=0.965$, $F1=0.959$. On Dominica (Table 2), FAR is reduced to 0.13 (at 90% recall) from 0.83 (MPS-CD [6]).

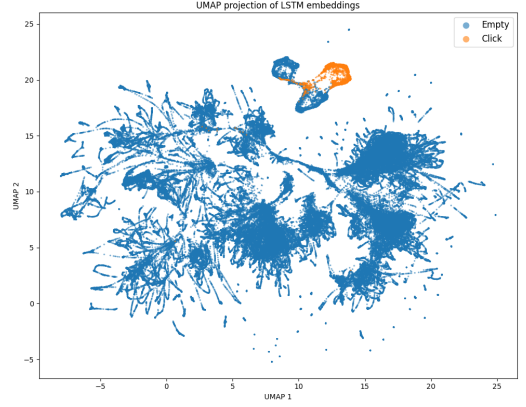


Figure 2: UMAP separates clicks (orange) from noise (blue).

Table 1: Evaluation of Metrics on Watkins Data at two tolerances. Best performing model is highlighted in bold.

Model	Params	0.5 ms tol.		20 ms tol.	
		<i>R</i>	<i>F</i> ₁	<i>R</i>	<i>F</i> ₁
SSCL [1]	Best from [1]	0.739	0.723	0.887	0.876
RI CNN	1-1-2	0.801	0.812	0.875	0.877
	1-2-1	0.875	0.877	0.965	0.973
	2-1-1	0.790	0.798	0.875	0.877
	2-2-2	0.857	0.862	0.875	0.877
BiLSTM	(5 × 32)	0.873	0.861	0.917	0.921
	(5 × 64)	0.947	0.942	0.973	0.989
	(5 × 128)	0.9652	0.959	0.991	0.995
	(10 × 32)	0.753	0.715	0.856	0.858
	(10 × 64)	0.958	0.945	0.966	0.987
	(10 × 128)	0.955	0.950	0.981	0.992

Table 2: False-alarm rate (FAR) on Dominica subset. Best performing model is highlighted in bold.

Detector	Pd @ R ≈ 0.90	FAR (10 s)
MPS-CD [6]	0.90	0.83
RI CNN	0.88	0.98
BiLSTM (5 × 128)	0.91	0.13

Our SSL+BiLSTM pipeline performs strongly and simplifies visual review. Interactive zoom and ms-aligned overlays reveal false alarms. Figure 2’s UMAP plot shows good separation. Future work will extend testing to live recordings from other marine species.

5 Conclusion

We deliver the *real-time, browser-based SSL detector* that reaches **99% F1** and reduces false alarms by **60%**. The drag-select UI, algorithm overlay, and live demo site turn a complex bioacoustic model into an actionable tool for multimedia researchers.

¹Source Code: <https://github.com/Animal-Speech-Understanding>.
Demo video: <https://drive.google.com/file/d/1d0urHU1rTVpawazetNT6PsgSqzD5fkYN>.
Slides from demo video: https://docs.google.com/presentation/d/170qNdezg0ZKNFe3FUu1dNb9vMdXik_R.

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